

■ Power BI

Top 100 Interview Questions

With Conversational Hinglish Answers

Freshers & Job-Seekers ke liye — Confidence ke saath
answer karo!

100 Questions • 6 Categories • Real Examples • Interview Tips

Accounts Expert • Power BI Course Notes

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How to Use This Guide

Isko effectively kaise padho aur practice karo

Step 1

Pehle poora question khud se answer karne ki koshish karo — phir answer dekho.

Step 2

Har answer ko apni language mein practice karo — ratta mat maaro, samjho aur bolo.

Step 3

Tips (■) specially important hain — yeh interviewer ko impress karne wale points hain.

Step 4

Conversational tone maintain karo — formal textbook jaisa mat bolo, natural raho.

Step 5

Real examples apne projects se add karo — aapka experience = strongest answer.

Step 6

Section 5 ke scenario questions sabse important hain — baar baar practice karo.

Section 1 — Power BI Basics

Q1–Q15 | Fundamentals jo hamesha puche jaate hain

Q1 Power BI kya hai aur yeh kyu use karte hain?

Power BI Microsoft ka Business Intelligence tool hai jo data ko connect karke, clean karke, model banake aur interactive dashboards/reports create karta hai. Hum isko isliye use karte hain kyunki isme Excel jaisi familiarity hai, lekin capabilities bahut zyada hain — real-time data, drag-drop visuals, automatic refresh sab milta hai. Basically raw data ko actionable insights mein convert karna — yahi Power BI ka kaam hai.

■ *Interview mein ek real example do: 'Maine sales dashboard banaya jisme management monthly revenue, top products aur region-wise performance ek hi page pe dekh sakti thi.'*

Q2 Power BI ke teen main components kaun se hain?

Power BI ke teen main parts hain. Pehla Power BI Desktop — yeh free Windows application hai jahan hum report aur dashboard design karte hain. Doosra Power BI Service — yeh cloud-based portal hai (app.powerbi.com) jahan reports publish karte hain aur team ke saath share karte hain. Teesra Power BI Mobile — iOS aur Android app hai jisse on-the-go reports dekh sakte hain. Desktop mein banao, Service pe publish karo, Mobile pe dekho — yahi workflow hai.

■ *Bata do ki aapne konsa use kiya hai: 'Maine mainly Desktop use kiya report banana ke liye.'*

Q3 Import Mode aur DirectQuery mein kya fark hai?

Import Mode mein data Power BI file mein copy ho jaata hai — iska matlab file size badh sakti hai lekin performance bahut fast hoti hai aur offline bhi kaam karta hai. Hum schedule refresh set kar sakte hain. DirectQuery mein data source pe hi rehta hai — har click pe live query jaati hai database ko. Yeh tab use karte hain jab data bahut bada ho ya real-time chahiye, lekin thoda slow hota hai aur kuch DAX functions kaam nahi karte. Beginner projects mein hamesha Import Mode use karna chahiye.

■ *Simple rule: Less than 1 million rows → Import. More than that ya real-time → DirectQuery.*

Q4 Power Query kya hota hai aur kab use karte hain?

Power Query Power BI ka ETL layer hai — yaani Extract, Transform, Load. Jab bhi data load karna ho, pehle Power Query mein jaake data clean karte hain. Jaise wrong data types fix karna, extra spaces remove karna, blank rows delete karna, columns rename karna — yeh sab Power Query mein hota hai. Jo bhi transformation karte hain woh 'Applied Steps' mein save hoti hain aur har refresh pe automatically apply hoti hain. Basically model mein clean data aaye isliye Power Query use karte hain.

■ *Ek important point: Power Query mein M language use hoti hai background mein, lekin mostly GUI se kaam hota hai — code likhna zaroori nahi.*

Q5 DAX kya hai? Power Query M Language se kaise alag hai?

DAX matlab Data Analysis Expressions — yeh Power BI ki formula language hai jo data model load hone ke BAAD kaam karta hai. Hum DAX se Measures aur Calculated Columns banate hain, jaise Total Revenue = SUM(Sales[Amount]). M Language Power Query mein use hoti hai — yeh data LOAD hone SE PEHLE kaam karti hai, cleaning aur transformation ke liye. Simple words mein: M = data cleaning ka language, DAX = business calculations ka language. Dono alag stages pe alag kaam karte hain.

■ *Memory trick: M = Massage (data ko shape karo), DAX = Do Analytics (calculations karo).*

Q6 Calculated Column aur Measure mein kya difference hai?

Calculated Column table mein ek nayi column add karta hai — har row ke liye alag value store hoti hai. Yeh Row Context mein evaluate hoti hai aur model mein physically stored hoti hai. Measure ek aggregation hai jo visual ke filter ke hisaab se calculate hoti hai — stored nahi hoti, on-demand calculate hoti hai. Simple example: 'Profit = Revenue - Cost' agar har order ka individual profit chahiye → Calculated Column. 'Total Profit' agar visual mein aggregate chahiye → Measure. Measures lightweight hain, Calculated Columns file size badhaate hain.

■ *Interview tip: 'Jab bhi aggregation (SUM, COUNT, AVG) chahiye → Measure. Jab row-level value chahiye → Calculated Column.'*

Q7 Star Schema kya hota hai? Power BI mein kyu important hai?

Star Schema ek data modeling pattern hai jisme ek central Fact table hoti hai jo multiple Dimension tables se connected hoti hai. Fact table mein transactions hote hain jaise orders, sales. Dimension tables mein context hota hai jaise Products, Customers, Date. Power BI mein yeh isliye important hai kyunki Star Schema se DAX queries fastest hoti hain, filter context correctly flow karta hai, aur reports simple aur accurate hoti hain. Hamesha Star Schema prefer karo — flat tables ya Snowflake schema se bachna chahiye.

■ *Visual explain karo: 'Star jaisi shape banti hai — centre mein Fact, bahar Dimensions — isliye Star Schema.'*

Q8 Relationship kaise banate hain Power BI mein? Kitne types hote hain?

Relationship banane ke liye Model View mein jaate hain aur ek table ke column ko doosri table ke column pe drag karte hain. Ya Home → Manage Relationships se bhi bana sakte hain. Types ki baat karein toh One-to-Many sabse common hai jisme ek customer ke many orders hote hain. One-to-One mein har row ek hi row se match hoti hai. Many-to-Many generally avoid karte hain kyunki ambiguous results aa sakte hain — iske liye bridge table use karna better hai. Cross-filter direction Single hona chahiye Star Schema mein.

■ *Always check karo ki '1' dimension side pe hai aur '*' fact side pe — yahi correct direction hai.*

Q9 Filter Context aur Row Context kya hota hai?

Yeh DAX ka sabse important concept hai. Row Context tab exist karta hai jab Calculated Column mein formula likhte hain — formula har row ke liye alag evaluate hoti hai, current row ki values use kar sakta hai. Filter Context tab exist karta hai jab Measure evaluate hota hai — visual mein jo slicers, filters, axes lagaye hain woh filter context banate hain. Jaise agar chart mein 'Electronics' filter lagaya hai toh Total Revenue measure sirf Electronics ke orders pe calculate hoga. CALCULATE function filter context change karne ke liye use hota hai.

■ *Easy trick: Row Context = table mein jaake row by row socho. Filter Context = report ka user kya dekh raha hai woh socho.*

**Q1
0****CALCULATE function kya karta hai? Ek example do.**

CALCULATE sabse important DAX function hai. Yeh kisi expression ko modified filter context mein evaluate karta hai. Matlab hum current slicer ya chart ke filter ko override ya add karke apna custom filter laga sakte hain. Example: Electronics ki sales sirf dikhani hai regardless of slicer — toh likhenge `CALCULATE([Total Revenue], Products[Category] = 'Electronics')`. Chahe report mein koi bhi filter laga ho, yeh measure hamesha Electronics ka hi revenue dikhayega. Basically CALCULATE = filter context change karo.

■ *Ek sentence mein: CALCULATE woh measure hai jo 'apni marzi ka filter' laata hai.*

**Q1
1****Power BI mein Date Table kyu banana zaroori hai? Kaise banate hain?**

Time Intelligence functions jaise TOTALYTD, SAMEPERIODLASTYEAR sirf tab kaam karte hain jab ek proper Date Table ho. Yeh table mein January 1 se December 31 tak continuous dates honi chahiye — koi gap nahi. Date Table banane ke liye DAX mein CALENDARAUTO() use karte hain jo automatically range detect karta hai. Phir Table Tools mein jaake 'Mark as Date Table' click karte hain. Bina yeh step ke Time Intelligence functions BLANK return karte hain.

■ *Common mistake: Date table banaya lekin 'Mark as Date Table' karna bhool gaye — yahi sabse common error hai.*

**Q1
2****Slicer aur Filter Pane mein kya difference hai?**

Slicer canvas pe visible hota hai — user directly click karke filter lagata hai. Yeh self-service filtering ke liye best hai aur non-technical users ke liye easy hai. Filter Pane report ke right side mein hota hai — developer aur advanced users ke liye. Filter Pane mein Visual level, Page level aur Report level teen types hote hain. Slicer mein sirf woh field filter hoti hai jo Slicer mein dali hai. Jab dashboard mein interactive buttons chahiye → Slicer. Jab developer side se filter lock karna ho → Filter Pane.

■ *Best practice: User ko Slicer do filtering ke liye, Filter Pane background mein default filters lagaane ke liye use karo.*

**Q1
3****Power BI mein Publish kaise karte hain? Service aur Desktop mein kya fark hai?**

Report publish karne ke liye Desktop mein Home tab → Publish click karte hain, phir workspace select karte hain. Service (app.powerbi.com) pe report appear ho jaati hai. Desktop mein report design hoti hai — .pbix file locally hoti hai. Service pe publish hone ke baad team ke saath share kar sakte hain, schedule refresh set kar sakte hain, aur mobile pe dekh sakte hain. Service pe jaake Datasets → Settings mein Scheduled Refresh set karte hain taaki data automatically update hota rahe.

■ Interview mein bolo: 'Desktop = development environment, Service = production/sharing environment.'

**Q1
4****Power BI mein Drill Down aur Drill Through mein kya fark hai?**

Drill Down ek hi visual ke andar hierarchy mein neeche jaana hai — jaise Year se Quarter, Quarter se Month. Visual pe double-click ya drill icon se kaam karta hai. Koi nayi page nahi khulti. Drill Through ek summary page se click karke ek alag detail page pe jaana hai — context filter ke saath. Jaise Region chart pe right-click → Drill Through → Region Details page. Yeh nayi page pe le jaata hai jo us specific item ke liye filtered hoti hai. Drill Down = same visual mein zoom. Drill Through = alag page pe jaao.

■ Simple: Drill Down = deep dive in same place. Drill Through = jump to new page with context.

**Q1
5****What is a KPI visual? Kab use karte hain?**

KPI visual teen cheezein ek saath dikhata hai: Actual value, Target value aur Trend. Jaise Sales Manager chahta hai ki actual monthly sales, target, aur arrow (green = achieved, red = missed) ek saath dikhe — yahi KPI visual karta hai. Fields mein Indicator (actual), Trend Axis (time), aur Target Goals (target measure) daale jaate hain. Bina target ke KPI ka koi matlab nahi — simple Card use karo. KPI tab best hai jab performance tracking chahiye with visual status.

■ Card = sirf number. KPI = number + target + trend. Yeh fark hamesha remember rakho.

Section 2 — DAX Functions

Q16–Q35 | Sabse common DAX interview questions

Q1
6**SUM aur SUMX mein kya difference hai?**

SUM simply ek column ki saari values add karta hai — `SUM(Sales[Revenue])`. SUMX ek iterator function hai jo table mein row by row jaata hai, har row pe ek expression calculate karta hai, phir sab results add karta hai. `SUMX(Sales, Sales[Qty] * Sales[Price])` — pehle har row mein $Qty \times Price$ calculate hoga, phir sab add. Jab ek column directly sum karna ho → SUM. Jab pehle per-row calculation karni ho phir sum karna ho → SUMX.

■ *Example: Revenue column exist karta hai → SUM use karo. Revenue column nahi hai but Qty aur Price hain → SUMX use karo.*

Q1
7**DIVIDE function kyu use karte hain simple / operator ke bajaye?**

Jab denominator 0 hota hai toh / operator se Error aata hai j■ report mein ugly dikhta hai. `DIVIDE(numerator, denominator, alternate)` safe hai — agar denominator 0 hota hai toh BLANK ya apna custom value return karta hai. Example: Profit Margin = `DIVIDE([Total Profit], [Total Revenue], 0)` — agar revenue 0 hai toh 0 return karega, error nahi. Yeh best practice hai — hamesha DIVIDE use karo division ke liye.

■ *Never write `[A] / [B]` in DAX. Always `DIVIDE([A], [B], 0)`. Ek habit bana lo.*

Q1
8**ALL, ALLEXCEPT aur ALLSELECTED mein kya difference hai?**

ALL ek table ya column se saare filters hata deta hai — % of Total calculate karne ke liye use hota hai. ALLEXCEPT ek table ke saare filters hata deta hai except jo columns specify karo — category total rakhke product filter hatane ke liye. ALLSELECTED visual filters hata deta hai lekin slicer selections rakhta hai — yeh slicer ke andar % calculate karne ke liye use hota hai. Teenon CALCULATE ke andar filter argument ke roop mein use hote hain.

■ *ALL = sab hatao. ALLEXCEPT = yeh raho baki sab hatao. ALLSELECTED = visual filter hatao slicer raho.*

**Q1
9****RELATED aur RELATEDTABLE mein kya fark hai?**

RELATED ek Calculated Column mein use hota hai — relationship ke through '1' side ki table se value fetch karta hai. Jaise Sales_Fact mein ProductID hai, RELATED(Products[Category]) se us row ke product ki category milegi. RELATEDTABLE uska ulta hai — '1' side ki table mein use karke '*' side ki related rows milti hain. Products table mein RELATEDTABLE(Sales_Fact) se us product ke saare orders milenge. Basically RELATED = ek value. RELATEDTABLE = ek table of rows.

■ Simple: *RELATED(column) → ek value return karta hai. RELATEDTABLE(table) → multiple rows return karta hai.*

**Q2
0****TOTALYTD, TOTALMTD aur TOTALQTD kya karte hain?**

Yeh Time Intelligence functions cumulative totals calculate karte hain. TOTALYTD January 1 se current date tak ka running total deta hai — har naye mahine mein pichla add hota rehta hai. TOTALMTD month ke pehle din se current date tak. TOTALQTD quarter start se current date tak. Teen cheezein zaroori hain: Date table hona, Mark as Date Table kiya hua hona, aur Date column se relationship hona. TOTALYTD(measure, 'Date'[Date]) — syntax yahi hai.

■ Test karo: Line chart mein TOTALYTD lagao aur dekho — line sirf upar jaaniti chahiye, kabhi neeche nahi.

**Q2
1****SAMEPERIODLASTYEAR kya karta hai? Kab use karte hain?**

SAMEPERIODLASTYEAR exactly same dates 1 year pehle ka data return karta hai. Jaise March 2025 ka data dekh rahe ho → SAMEPERIODLASTYEAR(Date[Date]) March 2024 ka data dega. Hum isse Year-over-Year comparison ke liye use karte hain. Formula: Sales LY = CALCULATE([Total Revenue], SAMEPERIODLASTYEAR('Date'[Date])). Phir YoY Growth % = DIVIDE([Total Revenue] - [Sales LY], [Sales LY], BLANK()).

■ 2024 months mein Sales LY = BLANK aayega agar 2023 ka data nahi hai dataset mein — yeh correct behavior hai.

**Q2
2****VAR / RETURN pattern kya hai aur kyu use karte hain?**

VAR intermediate calculation ko store karta hai ek naam se, RETURN final output specify karta hai. Faida yeh hai ki complex measures readable hoti hain, same value baar-baar calculate nahi hoti (performance better), aur debugging easy hoti hai. Example: VAR CurrentSales = [Total Revenue] VAR LastYear = [Sales LY] RETURN DIVIDE(CurrentSales - LastYear, LastYear, 0). Bina VAR ke LastYear formula do baar likhna padta — zyada code, slow calculation.

■ *Professional tip: 3 ya zyada sub-calculations wale har measure mein VAR use karo — yeh industry best practice hai.*

**Q2
3****FILTER function kab use karte hain? Simple condition se kab better hai?**

FILTER ek table return karta hai jisme sirf woh rows hoti hain jo condition satisfy karti hain. Jab condition column value pe based ho — Products[Category] = 'Electronics' — toh seedha CALCULATE mein likh sakte hain. FILTER tab use karte hain jab condition ek MEASURE pe based ho — jaise revenue > 5000 — kyunki measures column nahi hote. FILTER(Sales, [Total Revenue] > 5000) — aise likhte hain. Simple column condition → direct CALCULATE. Measure-based condition → FILTER zaruri hai.

■ *Performance tip: Simple column filter always FILTER se faster hota hai — isliye sirf zaroori ho tab hi FILTER use karo.*

**Q2
4****RANKX kaise kaam karta hai? Ek example do.**

RANKX kisi table ki values ko measure ke hisaab se rank karta hai. Syntax: RANKX(table, expression, value, order, ties). Example: Product Rank = RANKX(ALL(Products[ProductName]), [Total Revenue],, DESC, SKIP). ALL isliye use karte hain taaki visual filter ignore ho aur sab products rank hon. DESC matlab highest revenue = rank 1. SKIP matlab ties mein 1,2,2,4 — 3 skip hoga. DENSE lagao toh 1,2,2,3 aayega.

■ *Common mistake: ALL() nahi lagate toh sirf visible products rank honge — wrong result. Hamesha ALL() use karo RANKX mein.*

**Q2
5****LOOKUPVALUE kya karta hai? RELATED se kab better hai?**

LOOKUPVALUE bina relationship ke ek table se value fetch karta hai. LOOKUPVALUE(result_col, search_col, search_value) — jaise VLOOKUP jaisa. RELATED tab use karo jab active relationship hai — faster aur simpler hai. LOOKUPVALUE tab use karo jab relationship banana possible nahi — jaise different granularity hai ya legacy data hai. Example: Price = LOOKUPVALUE(Products[Price], Products[ProductID], Sales[ProductID]) — relationship nahi hai fir bhi price milega.

■ *Always prefer RELATED over LOOKUPVALUE agar relationship exist karti hai — RELATED zyada efficient hai.*

**Q2
6****IF aur SWITCH mein kya difference hai? SWITCH(TRUE()) kya hota hai?**

IF simple ek condition check karta hai — agar true toh A, warna B. Nested IF multiple conditions ke liye but bahut unreadable hota hai. SWITCH multiple exact values check karta hai — cleaner syntax. SWITCH(TRUE()) sabse powerful pattern hai jisme har condition independently check hoti hai — ranges ke liye best. Example: SWITCH(TRUE(), [Revenue]>10000, 'Gold', [Revenue]>5000, 'Silver', 'Bronze'). Yeh nested IF ka clean replacement hai.

■ *Rule: 2 conditions → IF. 3-4 exact values → SWITCH. Range conditions → SWITCH(TRUE()).*

**Q2
7****SELECTEDVALUE aur HASONEVALUE kya karte hain?**

SELECTEDVALUE slicer se selected value return karta hai — agar ek value selected hai toh woh value, warna default value. Example: SELECTEDVALUE('Date'[Year], 'All Years') — agar 2025 selected hai toh 2025, koi selection nahi hai toh 'All Years'. HASONEVALUE TRUE/FALSE return karta hai — ek value select hai ya nahi. Yeh dynamic titles banane ke liye use karte hain: Chart Title = 'Revenue — ' & SELECTEDVALUE(Products[Category], 'All Categories').

■ *Use case: Dynamic report title jo automatically slicer selection ke saath change hota hai — very professional!*

**Q2
8****COUNTROWS aur COUNT mein kya difference hai?**

COUNT sirf numeric values count karta hai — blank aur text ignore karta hai. COUNTA koi bhi non-blank value count karta hai. COUNTROWS table ki total rows count karta hai — kisi column ki zaroorat nahi. For orders: Total Orders = COUNTROWS(Sales_Fact) — sabse accurate hai. COUNT(Sales_Fact[OrderID]) bhi same result dega agar OrderID column mein koi blank nahi hai. Lekin COUNTROWS more reliable hai aur COUNTROWS(FILTER(...)) pattern bahut use hota hai.

■ *Best practice: Orders count ke liye hamesha COUNTROWS use karo — COUNT se errors ho sakte hain agar column mein blanks hain.*

**Q2
9****DISTINCTCOUNT kab use karte hain?**

DISTINCTCOUNT unique values ki count karta hai — duplicates ignore karta hai. Jaise ek customer ne 10 orders diye hain — COUNT 10 dega, DISTINCTCOUNT(Sales[CustomerID]) unique customers count karega. Use case: Unique Customers = DISTINCTCOUNT(Sales[CustomerID]). Unique Products sold = DISTINCTCOUNT(Sales[ProductID]). Yeh Customer analytics mein bahut important hai — total transactions aur unique customers alag hote hain.

■ *Interview example: 'E-commerce mein repeat customers track karna zaroori hai — DISTINCTCOUNT se unique buyers nikalte hain.'*

**Q3
0****SUMMARIZE aur ADDCOLUMNS kab use karte hain?**

SUMMARIZE ek new table banata hai grouped data ke saath — SQL GROUP BY jaisa. SUMMARIZE(Sales, Products[Category], 'Total', [Total Revenue]) — category-wise revenue ki table milegi. ADDCOLUMNS ek existing table mein computed columns add karta hai without losing original rows. ADDCOLUMNS(Products, 'Revenue', [Total Revenue]) — sab products ke saath unka revenue. Yeh calculated tables banane ke liye use hote hain — Model View mein Modeling → New Table se likhte hain.

■ *Practical use: Report ke liye ek pre-calculated table chahiye toh SUMMARIZE. Existing table mein columns add karni hain toh ADDCOLUMNS.*

Q3
1**IFERROR aur ISBLANK kab use karte hain?**

IFERROR expression execute karta hai, agar error aaye toh alternate value deta hai. Jaise `IFERROR(DIVIDE([A],[B]), 0)` — agar divide mein error hai toh 0. ISBLANK check karta hai ki value blank hai ya nahi — `TRUE/FALSE` return karta hai. Jaise `IF(ISBLANK([Total Revenue]), 'No Data', [Total Revenue])`. Yeh error handling ke liye use hote hain — reports mein ugly error messages na dikhein.

■ *Best practice: Har DIVIDE mein third argument as alternate result zaroor do — IFERROR wrapper ki zaroorat nahi padegi.*

Q3
2**FORMAT function kab use karte hain? Example do.**

FORMAT number ya date ko formatted string mein convert karta hai. Revenue formatted as currency: `FORMAT([Total Revenue], '■#,##0.00')`. Date formatted as month name: `FORMAT(Sales[OrderDate], 'MMMM YYYY')` — 'January 2025'. Day name: `FORMAT(Sales[OrderDate], 'dddd')` — 'Monday'. Yeh display purposes ke liye use hota hai — calculated columns mein jab human-readable text chahiye. Lekin numeric formatting ke liye Modeling tab se Format set karna better hai measures ke liye.

■ *Warning: FORMAT string return karta hai — iske baad math operations nahi kar sakte. Sirf display ke liye use karo.*

Q3
3**USERELATIONSHIP kab use karte hain?**

Jab do tables ke beech ek se zyada relationships hoon toh sirf ek active hoti hai. Baaki inactive relationships ko measure mein activate karne ke liye USERELATIONSHIP use karte hain. Example: Orders table mein OrderDate aur ShipDate dono Date table se connect hain — lekin sirf OrderDate active hai. Ship date pe analysis ke liye: `CALCULATE([Revenue], USERELATIONSHIP(Orders[ShipDate], 'Date'[Date]))`. Yeh CALCULATE ke andar filter argument ki jagah likhte hain.

■ *Use case: Delivery analysis ke liye ShipDate aur Sales analysis ke liye OrderDate — USERELATIONSHIP se dono possible hain.*

Q3
4**Percentage of Total kaise calculate karte hain DAX mein?**

ALL() function use karte hain jo sab filters remove karta hai. Formula: $\text{Revenue \% Total} = \text{DIVIDE}([\text{Total Revenue}], \text{CALCULATE}([\text{Total Revenue}], \text{ALL}(\text{Sales_Fact})), 0)$. Jab yeh measure Table visual mein ProductName ke saath lagate hain toh har product apna % share dikhata hai. ALL(Sales_Fact) se sirf Sales_Fact ke filters hate hain — agar Products dimension pe filter hai woh bhi hata do — ALL(Products) ya ALL() use karo grand total ke liye.

■ *Test karo: Sab products ke % sum hone chahiye 100% — agar nahi ho raha toh ALL() sahi jagah nahi laga.*

Q3
5**Running Total (Cumulative Sum) kaise banate hain?**

Running Total ke liye FILTER aur ALL combination use karte hain: $\text{Running Total} = \text{CALCULATE}([\text{Total Revenue}], \text{FILTER}(\text{ALL}('Date'), 'Date'[Date] \leq \text{MAX}('Date'[Date])))$. Yeh current date tak ki saari dates include karta hai — cumulative sum ban jaata hai. Ya TOTALYTD use karo agar year-based cumulative chahiye. Running Total sirf date se related nahi bhi ho sakta — jaise cumulative orders by product rank bhi bana sakte hain.

■ *TOTALYTD simple hai but sirf Jan-Dec reset karta hai. Custom Running Total zyada flexible hai.*

Section 3 — Visuals & Report Design

Q36–Q55 | Visualization ke questions

Q3
6**Kaun sa visual kab use karte hain? Basic rule kya hai?**

Category comparison → Bar ya Column Chart. Time trend → Line Chart. Part-of-whole → Pie/Donut (sirf 2-5 categories). Hierarchy → Treemap. Correlation → Scatter Chart. Single KPI → Card. Target vs Actual → KPI visual. Detailed data → Table ya Matrix. Geographic → Map. Process drop-off → Funnel. Step-by-step change → Waterfall. Basic rule: Jo story sunani hai woh visual choose karo — chart ka style data ki nature se match hona chahiye.

■ *Interview tip: 'Wrong visual sahi data bhi galat story sunata hai.' Example do: Revenue trend ke liye Pie Chart mat use karo.*

Q3
7

Clustered Column aur Stacked Column mein kya fark hai?

Clustered Column mein categories side-by-side bars mein hote hain — individual comparison easy hoti hai. Stacked Column mein sub-categories ek hi bar mein stack hoti hain — total aur breakdown ek saath dikhta hai. Clustered: 'A vs B vs C kitna bada hai' samjhana ho. Stacked: 'Total mein A + B + C ka share' dikhana ho. 100% Stacked sirf percentage dikhata hai, actual values nahi. 5 se zyada segments ek bar mein mat stack karo — unreadable ho jaata hai.

■ *Simple: Comparison chahiye → Clustered. Contribution chahiye → Stacked. Mix percentage chahiye → 100% Stacked.*

Q3
8

Line Chart aur Bar Chart mein kab kaunsa choose karein?

Line Chart time series ke liye best hai — X axis pe dates/months hote hain, continuous trend dikhta hai. Bar Chart category comparison ke liye — discrete items compare karte hain. Agar X axis pe January, February, March hain aur trend dikhana hai → Line Chart. Agar North, South, East, West regions compare karni hain → Bar Chart. Important: Bar Chart pe time dalna wrong practice hai — Line Chart use karo time ke liye.

■ *Test: X axis pe time hai? → Line Chart. X axis pe categories hain? → Bar/Column Chart.*

Q3
9

Matrix aur Table mein kya difference hai?

Table flat, row-by-row data dikhata hai — ek hi dimension ke saath. Transaction list, customer details — Table ke liye best hai. Matrix pivot table jaisa hai — Rows aur Columns dono pe categories hoti hain. Region × Month × Revenue ek Matrix mein aayega. Matrix mein drill-down bhi hota hai — Category expand karo Products dikhenge. Conditional formatting dono mein available hai. Jab do-dimensional cross-tab analysis chahiye → Matrix. Jab line-by-line detail chahiye → Table.

■ *Real example: Accountant ko har invoice → Table. Finance manager ko Region × Quarter analysis → Matrix.*

**Q4
0****Conditional Formatting kaise use karte hain visuals mein?**

Table ya Matrix mein kisi column pe right-click → Conditional Formatting. Types: Background Color (heatmap — low=white, high=blue), Font Color (red=negative, green=positive), Data Bars (mini bar charts inside cells), Icons (traffic lights based on rules). Color Scale automatically highest aur lowest pe based color lagata hai. Rules-based mein aap decide karte ho: 'If > 40% then green'. Yeh data patterns instantly visible karta hai bina chart ke.

■ *Pro tip: Revenue column mein Data Bars lagao — ek glance mein relative performance dikh jaata hai. Very professional look!*

**Q4
1****Drill Through kaise setup karte hain?**

Pehle ek detail page banao — jaise 'Product Detail'. Us page ke Filters pane mein Drill Through filter area mein ProductName drag karo. Ab main page pe kisi product bar pe right-click karo → Drill Through → Product Detail. Automatically us product ke liye filtered page khulega. Detail page mein Back button zaroor add karo — Insert → Buttons → Back. Page navigation se hide karna best practice hai — Format → Page information → Hide page.

■ *Common mistake: Drill Through filter field target page pe nahi dala — bina iske right-click option nahi aayega.*

**Q4
2****Tooltip Page kya hota hai? Kaise banate hain?**

Tooltip Page ek custom mini-report hai jo hover karne pe popup hota hai — default tooltip se zyada rich information de sakta hai. Banane ke liye: Naya page add karo → Format → Page Information → Allow use as tooltip: ON → Page size Tooltip type set hoti hai. Is page pe visuals design karo. Main chart pe jaake Format → Tooltips → Type: Report Page → apna tooltip page select karo. Ab hover karte hi mini dashboard popup hoga.

■ *Best practice: Tooltip page mein 2-3 small cards ya ek mini chart rakho — bahut zyada information confusing lagti hai.*

**Q4
3****Bookmarks kya hote hain? Kab use karte hain?**

Bookmarks report ki current state save karte hain — filters, slicer selections, visual visibility sab capture hota hai. Ek click se saved state mein vaapas aa sakte hain. Use case: 'Q1 View' aur 'Q2 View' bookmarks banao — buttons se toggle karo. Ya specific filter set save karo presentation ke liye. View tab → Bookmarks pane → Add se create karte hain. Buttons ke saath link karte hain: Button Format → Action → Bookmark type → select bookmark.

■ *Storytelling ke liye perfect: Presentation mein ek click pe specific view dikhao — bina manually filter change kiye.*

**Q4
4****Pie Chart kab use nahi karna chahiye?**

Pie Chart sirf 2-5 categories ke liye suitable hai. Zyada categories mein slices itni chhoti ho jaati hain ki compare karna impossible hota hai. Precise comparison ke liye Pie Chart weak hai — adjacent slices judge karna difficult hota hai. Time trend ke liye Pie Chart kabhi use mat karo. Agar 'kaunsa segment sabse bada hai' exactly compare karna hai → Bar Chart better hai. Pie Chart sirf 'roughly kitna proportion hai' dikhane ke liye theek hai.

■ *Rule: 5 se zyada categories hain? Pie mat uso. Exact comparison chahiye? Bar Chart use karo. Time data? Line Chart use karo.*

**Q4
5****Q&A; Visual aur Smart Narrative kya karte hain?**

Q&A; Visual natural language mein question type karne pe automatic chart generate karta hai. 'Show total sales by region' type karo → automatically bar chart ban jaata hai. Non-technical users ke liye perfect self-service tool. Smart Narrative AI se text summary generate karta hai — 'Sales 15% badhi, North region leader raha' jaisa paragraph automatically. Dono Insert tab se add hote hain. Q&A; ke liye field synonyms set karna important hai better recognition ke liye.

■ *Interview point: 'Yeh AI-powered features Power BI ko sirf visualization tool se intelligence tool banate hain.'*

Q4
6

Slicer sync across pages kaise karte hain?

View tab → Sync Slicers. Ek pane aayegi jisme sab pages list honge. Jis page pe slicer visible aur synced banana ho woh dono columns tick karo — 'Sync' aur 'Visible'. Ab ek page pe slicer change karo — synced pages pe automatically same filter lagega. Example: Date Range slicer Page 1 pe hai lekin Page 2 aur 3 pe bhi same filter chahiye — Sync Slicers se ek hi slicer control sab pages karta hai.

■ *Common use case: Executive summary page aur detail pages dono ek hi date slicer se control karein.*

Q4
7

Waterfall Chart kab use karte hain?

Waterfall Chart step-by-step increase aur decrease dikhata hai — starting value se ending value tak kaise pohuncha. Best use: Profit Bridge (Opening profit → Revenue add → Expenses subtract → Closing profit), Budget Variance, Headcount changes. Increase green, Decrease red, Total/Subtotal grey hota hai. Setup mein Category mein steps dalo, Y Axis mein values dalo, aur Total columns ko 'Is subtotal' mark karo. Accountants aur Finance professionals ke liye very useful visual.

■ *Common mistake: Total column ko regular bar ki tarah rakhna — isko 'Is subtotal' mark karo warna chart galat dikhega.*

Q4
8

Map visual kaise use karte hain? Kya chahiye data mein?

Map visual geographic data dikhata hai. Data mein City, State, Country ya Latitude/Longitude columns hone chahiye. Filled Map mein regions color-coded hote hain — Sales by State. Bubble Map mein cities pe bubbles hote hain — size = value. Power BI automatically location detect karta hai text se (Mumbai, Maharashtra, India). Agar ambiguity ho toh Location field mein Country level bhi dalo. Azure Maps custom styling ke liye available hai.

■ *Important: Bing Maps ke liye internet connection chahiye. Offline mode mein maps blank dikhenge.*

**Q4
9**

Report Theme kya hota hai? Kaise apply karte hain?

Report Theme sab pages pe consistent colors, fonts aur visual formatting apply karta hai. View tab → Themes → built-in themes available hain (Executive, City Park etc). Custom theme JSON file se apply kar sakte hain — View → Themes → Browse for themes. Brand colors match karne ke liye custom theme best practice hai. Theme change karte hi saare visuals automatically update ho jaate hain — manually har visual format nahi karna padta.

■ *Professional tip: Client project mein hamesha client ki brand colors se custom theme banao — very impressive presentation hoti hai.*

**Q5
0**

Page Navigation buttons kaise banate hain?

Insert tab → Buttons → Navigator se built-in page navigator add hota hai. Ya custom buttons: Insert → Buttons → Blank → Format → Action → Page Navigation → Destination page select karo. Button text, icon, hover state customize karo. Best practice: Sabhi pages pe same navigation panel rakho. Page jis pe ho uska button darker ya active state dikhe. Yeh non-technical users ke liye report experience bahut improve karta hai.

■ *Pro tip: Selection pane use karo — visuals show/hide bookmarks se toggle karo — advanced interactivity banao.*

**Q5
1**

Gauge Chart kab use karte hain? Limitations kya hain?

Gauge Chart ek single KPI ka target ke against progress speedometer style mein dikhata hai. Min value, max value aur target set karte hain — needle current value dikhata hai. Jab ek metric ka single snapshot chahiye with min-max range. Limitation: Space bohot leta hai sirf ek value ke liye. Multiple values compare nahi kar sakta. Dashboard mein space ki constraint ho toh Card ya KPI prefer karo. Best use: Monthly target achievement screen pe real-time dikhana.

■ *When to use: Executive dashboard mein single most important metric prominently dikhana ho.*

Q5
2

Scatter Chart aur Bubble Chart mein kya difference hai?

Scatter Chart do numeric variables ke beech relationship dikhata hai — X axis pe ek metric, Y axis pe doosri. Correlation positive ya negative dikh ta hai. Bubble Chart mein teesri dimension add hoti hai — bubble ki size ek numeric value represent karta hai. Marketing Spend vs Sales (Scatter) + bubble size = Customer Count (Bubble). Dono ke liye data mein continuous numeric values chahiye — categorical data nahi chalta.

■ *Correlation ≠ Causation: Scatter Chart correlation dikhata hai — causation prove karne ke liye alag analysis chahiye. Interview mein yeh point important hai.*

Q5
3

Funnel Chart kab use karte hain?

Funnel Chart sequential stages mein drop-off rate dikhata hai. Sales pipeline: Leads → Contacted → Proposal → Demo → Closed. Hiring funnel: Applications → Screened → Interview → Offer → Joined. Har stage se agla stage mein kitna % conversion hua woh clearly dikhta hai. Stages ka natural sequence hona zaroori hai — random categories ke liye Bar Chart better hai. Data labels mein % of first stage dikhana insight zyada deta hai.

■ *Business value: 'Proposal se Closing mein 70% drop — sales pitch improve karo.' Data se actionable insight!*

Q5
4

Treemap kab better hota hai Pie Chart se?

Treemap hierarchical data ke liye perfect hai — Category → SubCategory → Product sab ek visual mein. Rectangle ka size value represent karta hai. Pie Chart flat data ke liye hai — hierarchy nahi. Treemap mein 10-15 categories bhi readable hoti hain (small boxes clearly visible). Pie mein 5 se zyada slices unreadable ho jaate hain. Proportion + hierarchy ek saath dikhana ho → Treemap. Simple proportion → Pie/Donut.

■ *Interview example: 'Product category → subcategory → individual product revenue ek visual mein → Treemap perfect choice hai.'*

**Q5
5****Ribbon Chart kab use karte hain?**

Ribbon Chart time ke saath categories ki ranking kaise badli dikhata hai. January mein Product A #1 tha, March mein Product B ne overtake kar liya — yeh crossing ribbons clearly dikhega. Jab sirf values track karni hoon → Line Chart. Jab rankings track karni hoon → Ribbon Chart. Example: Monthly top-selling products ki ranking changes track karna. 5 se zyada categories mein ribbons overlap ho jaate hain — readability kharab hoti hai.

■ *Unique use case: Market share shifts, leaderboard changes, ranking competitions — Ribbon Chart inke liye perfect hai.*

Section 4 — Data Modeling

Q56–Q70 | Model design ke important questions

**Q5
6****Many-to-Many relationship kab aati hai? Kaise handle karte hain?**

Many-to-Many tab aati hai jab dono tables mein same key ki duplicate values hoti hain — jaise ek order ke multiple products, ek product multiple orders mein. Power BI direct M:M support karta hai but results ambiguous ho sakte hain. Best practice: Bridge/Junction table use karo. Orders aur Products ke beech OrderItems table — OrderID aur ProductID dono columns. Ab Orders → OrderItems (1:M) aur Products → OrderItems (1:M) — clean Star Schema ban jaata hai.

■ *Interview answer: 'Many-to-Many ko directly avoid karo — always bridge table use karo for clean relationships.'*

Q5
7**Inactive relationship kya hoti hai? Kab create hoti hai?**

Do tables ke beech sirf ek active relationship ho sakti hai. Agar second relationship add karo toh woh inactive ho jaati hai — Model View mein dashed line se dikhti hai. Jaise Orders table mein OrderDate aur ShipDate dono Date table se connect hain — ek active, ek inactive. Inactive relationship ko measure mein use karne ke liye USERELATIONSHIP() function use karte hain CALCULATE ke andar. Yeh multiple date analysis ke liye common pattern hai.

■ *Practical: Delivery analysis ke liye ShipDate se, Sales analysis ke liye OrderDate se — dono analyze kar sakte hain alag measures se.*

Q5
8**Cross-filter direction 'Both' kab use karte hain?**

Default Single direction mein filter sirf Dimension → Fact flow karta hai. 'Both' direction mein dono tables ek doosre ko filter kar sakti hain. 'Both' use karo: Many-to-Many relationships mein, Role-playing dimensions mein. Lekin hamesha 'Both' mat use karo — circular dependencies create ho sakti hain, performance slow hoti hai aur unexpected results aa sakte hain. Rule: Star Schema mein hamesha Single. Special cases mein carefully Both.

■ *Best practice: 'Both' use karne se pehle socho kya really zaruri hai — mostly Single direction hi kafi hota hai.*

Q5
9**Snowflake Schema aur Star Schema mein kya fark hai?**

Star Schema mein Dimension tables directly Fact table se connected hoti hain — one hop. Snowflake Schema mein Dimension tables further normalize hoti hain — multiple hops. Jaise Products → Subcategory → Category alag tables mein. Power BI mein Star Schema recommend kiya jaata hai kyunki query performance better hoti hai aur DAX simpler hota hai. Snowflake se complex relationships aur slow queries hoti hain. Real world mein normalize database se data load karo lekin Power BI mein denormalize karke Star Schema banao.

■ *Simple rule: Jo data aaya woh source ka — Power BI model mein hamesha Star Schema banana best practice.*

Q6
0

Role-Playing Dimensions kya hoti hain?

Jab ek dimension table multiple roles play kare — jaise Date table ko OrderDate ke liye bhi use karna aur ShipDate ke liye bhi. Date table ek hi hai lekin Orders table se do jagah connect hoti hai — ek active, ek inactive. Inactive relationship ko USERELATIONSHIP() se measures mein activate karte hain. Role-playing dimension ki wajah se table duplicate nahi karni padti — same Date table do roles play karta hai — OrderDate aur ShipDate dono analyze kar sakte hain.

■ *Example: Date_Dim → Orders_Fact[OrderDate] (active) aur Date_Dim → Orders_Fact[ShipDate] (inactive).*

Q6
1

Data Model mein unused columns aur tables rakhna chahiye ya nahi?

Nahi — unused columns aur tables remove karne chahiye. Har column jo model mein load hoti hai memory occupy karti hai — file size aur refresh time badh jaate hain. Power Query mein jo columns use nahi honge unhe 'Choose Columns' se remove karo loading se pehle. Model View mein jo tables use nahi ho rahi unhe 'Hide' karo ya remove karo. Yeh performance optimization ka important step hai — especially bade datasets mein.

■ *Best practice: 'Load sirf woh columns jinhe report mein use karna hai — baaki sab Power Query mein hi filter kar do.'*

Q6
2

Calculated Table kya hota hai? Kab use karte hain?

Calculated Table DAX expression se banaya jaata hai — Modeling → New Table. Example: Date Table banane ke liye CALENDARAUTO() use karte hain. Ya SUMMARIZE se category-wise summary table. Calculated tables model mein permanently stored rehti hain — memory use karta hai. Inka use: Date table, Bridge tables, aur special analysis tables ke liye. Power Query tables se alag — yeh DAX se data model ke andar banate hain source se nahi.

■ *Most common calculated table: Date Table using CALENDARAUTO() — yeh har project mein use hoti hai.*

**Q6
3****Bidirectional filtering ke risks kya hain?**

Bidirectional filtering yani Both direction cross-filter se kuch risks hain. Circular dependency create ho sakti hai — $A \rightarrow B$ aur $B \rightarrow A$ filter kare toh infinite loop jaisa. Ambiguous paths mein Power BI error deta hai. Performance slow hoti hai kyunki more calculations hoti hain. Unexpected results aa sakte hain — ek table ka filter doosri jagah unintentionally propagate ho. Isliye default Single direction rakho aur Both sirf specific jaroorat pe carefully use karo.

■ *Test: Both direction lagate hi agar report mein unexpected changes aayen → single pe wapas jao.*

**Q6
4****Power BI mein Aggregation Tables kya hote hain?**

Aggregation Tables pre-calculated summary data hoti hain jo detailed data ke saath rakhi jaati hain. Jab data bahut bada ho — million rows — toh har query pe aggregate karna slow hota hai. Aggregation Table mein pehle se City level, Category level summarized data hota hai. Power BI pehle Aggregation Table check karta hai — match hone par wahan se fast answer deta hai, otherwise detail table se. Beginner projects mein zaruri nahi — advanced performance optimization ke liye hota hai.

■ *Simple: Aggregation Tables = pre-calculated shortcuts for faster queries on big data.*

**Q6
5****Data Model mein Table aur Column rename karna kyu important hai?**

Report users aur fields pane mein table aur column names dikhte hain. 'tbl_fct_orders_2024' ki jagah 'Orders' zyada clean aur professional dikhta hai. DAX measures bhi readable hote hain — `SUM(Orders[Revenue])` is simpler than `SUM(tbl_fct_orders_2024[revenue_amt])`. Q&A; visual bhi better kaam karta hai readable names se. Power Query mein columns rename karo ya Model View mein properties se. Consistent naming convention rakho — Title Case prefer karo.

■ *Best practice: Table names singular rakho (Product not Products) ya plural choose karo but consistent raho throughout.*

**Q6
6**

What is a Measure Table? Kyu banate hain?

Measure Table ek empty table hoti hai sirf measures organize karne ke liye — koi data nahi hota. Jab measures 4-5 se zyada ho jaayein tab saari measures ek dedicated 'Measures' table mein rakho — clean organization ke liye. Enter Data se blank table banao, naam do '_Measures', phir saari measures yahan move karo. Fields pane mein sab measures ek jagah organize rehte hain. Yeh industry best practice hai large projects mein.

■ *Pro tip: Measure table ka naam underscore se shuru karo (_Measures) taaki Fields pane mein upar aaye — easy navigation.*

**Q6
7**

Slowly Changing Dimensions (SCD) kya hoti hain?

SCD dimension data hai jo time ke saath change hota hai. Jaise customer address change ho ya product price change ho. Type 1: Directly overwrite karo — purana data gone. Type 2: New row add karo with start/end date — history preserved. Power BI mein SCD Type 2 handle karne ke liye USERRELATIONSHIP ya date-filtered relationships use karte hain. Beginner level mein yeh advanced topic hai — mostly enterprise projects mein aata hai.

■ *Interview tip: Is concept ka naam pata hona impressive hai even if detailed implementation na aaye — 'SCD Type 1 and 2 ka concept jaanta hoon' bolna enough hai fresher ke liye.*

**Q6
8**

Power BI mein Row Level Security (RLS) kya hota hai?

RLS se different users ko alag-alag data dikhate hain same report mein — data security ke liye. Jaise North Region Manager sirf North ka data dekhe, South Manager sirf South ka. Roles banate hain Modeling tab mein, DAX filter add karte hain — [Region] = USERPRINCIPALNAME(). Service pe publish karne ke baad roles pe users assign karte hain. Static RLS simple hai, Dynamic RLS zyada scalable aur powerful hai.

■ *Important: RLS sirf Service aur mobile pe enforce hota hai — Desktop mein aap khud View As Role se test kar sakte hain.*

**Q6
9****Composite Model kya hota hai?**

Composite Model mein Import aur DirectQuery dono mode ka data ek hi report mein hota hai. Jaise historical data Import mode mein (fast), real-time data DirectQuery mein (live). Aggregation tables bhi Composite Model ka part hoti hain. Yeh advanced scenario hai jab single mode se requirements meet nahi hoti. Limitation: Kuch DAX functions restricted hain, performance carefully manage karna padta hai.

■ *Fresher interview mein: Concept jaanna enough hai — practical implementation enterprise level pe hoti hai.*

**Q7
0****Power BI file size kaise reduce karte hain?**

File size reduce karne ke liye: (1) Unused columns aur tables remove karo Power Query mein. (2) High cardinality columns (IDs with millions unique values) remove karo agar visual mein use nahi. (3) String columns compress nahi hote achhe se — integers prefer karo where possible. (4) Calculated tables kam rakho. (5) Import mode mein sirf zaruri data load karo — filters lagao Power Query mein. (6) Date/Time ko Date type mein rakho — Time alag column banao agar needed.

■ *Quick win: Power Query mein 'Choose Columns' use karo — sirf woh columns lo jo report mein use hongi. 40-50% size reduction ho sakta hai.*

Section 5 — Advanced & Scenario Questions

Q71–Q90 | Problem-solving aur scenario-based questions

**Q7
1****Sales Last Year BLANK aa raha hai — kya problem ho sakti hai?**

Teen reasons ho sakte hain. Pehla: Date table 'Mark as Date Table' nahi kiya — yeh sabse common reason hai. Doosra: Prior year ka data dataset mein hi nahi hai — jaise sirf 2024-2025 ka data hai toh 2024 mein previous year blank aayega. Teesra: Date table ki date range complete nahi — 2023 mein data hai but Date_Dim mein 2023 nahi — Time Intelligence fail karega. Solution: Date_Dim mark karo, data range check karo, aur relationship verify karo.

■ *Debug sequence: Mark as Date Table → Date range → Relationship check → Data availability check. Isi order mein diagnose karo.*

**Q7
2****Circular Dependency error kab aata hai? Kaise fix karte hain?**

Circular Dependency tab aata hai jab A, B pe depend karta hai aur B, A pe — loop ban jaata hai. Power BI yeh detect karta hai aur error deta hai. Common causes: (1) Bidirectional relationships + calculated tables combination. (2) Calculated column jo doosri calculated column ko reference kare jo pehli ko reference kare. Fix: Relationship direction Single rakho. Circular columns ko measures mein convert karo. Ya intermediate table banao loop todne ke liye.

■ *Prevention: Star Schema mein Single direction relationships rakhne se 90% circular dependency issues avoid ho jaate hain.*

**Q7
3****Report bahut slow load ho rahi hai — kya steps loge optimize karne ke liye?**

Step by step diagnosis: Pehle View → Performance Analyzer run karo — har visual ki query time dekho. Sabse slow visual identify karo. DAX query optimize karo — complex measures simplify karo, VAR use karo. Visuals ki count reduce karo — ek page pe 10+ visuals slow karte hain. Unused columns remove karo model se. High cardinality fields visual mein use kar rahe ho toh alternatives socho. Bidirectional relationships single karo. Large tables ke liye aggregation consider karo.

■ *Quick wins: (1) Visuals kam karo. (2) Performance Analyzer se slowest visual pakdo. (3) Unused columns hatao. Yeh teen steps 50% improvement de sakte hain.*

**Q7
4****Ek client kehta hai 'Dashboard mein live data chahiye' — aap kya karoge?**

Pehle clarify karo 'live' ka matlab — real-time ya near-real-time? Real-time: DirectQuery mode use karo — har interaction pe live query jaati hai source database pe. Near-real-time: Import mode mein scheduled refresh lagao — minimum Power BI Pro mein 8 refreshes per day, Premium mein 48 refreshes tak. Agar source streaming data hai — Power BI Streaming dataset use kar sakte hain for truly real-time dashboards. Tradeoff explain karo: Real-time = slower performance. Near-real-time = faster but slight delay.

■ *Always ask: 'Kitna delay acceptable hai?' Often 30-min refresh near-real-time enough hota hai — DirectQuery ki complexity avoid hoti hai.*

**Q7
5****Do Fact tables hain — inhe kaise model karo?**

Do Fact tables ko directly connect karna galat hai — Many-to-Many ya ambiguous results denge. Sahi approach: Star Schema mein dono Fact tables apni apni Dimension tables se connect karo. Agar ek common dimension hai jaise Products_Dim — dono fact tables separate connections se Products_Dim se connect karenge. Measures alag-alag banao dono tables ke liye. Cross-table analysis ke liye shared dimension bridge ka kaam karta hai — CALCULATE automatically dono tables se data combine karta hai.

■ *Remember: 2 Fact tables → 2 separate star schemas sharing common dimensions. Never connect fact to fact directly.*

**Q7
6****User report mein ek specific date range select karne pe BLANK dekh raha hai — kya reason ho sakta hai?**

Multiple reasons check karo: (1) Selected date range mein data hi nahi hai — sales nahi hui us period mein. (2) Date relationship galat column se bani hai — OrderDate ki jagah ShipDate se connect hua hai. (3) Date_Dim mein woh dates exist nahi karti — Date_Dim ki range orders ki dates cover nahi karti. (4) Slicer Date_Dim[Date] pe hai lekin measure Orders[OrderDate] use karta hai — mismatch. Har ek systematically check karo.

■ *Debug tip: Temporary ek Card visual add karo [Total Orders] ka — agar yeh bhi blank hai toh data issue. Agar yeh number dikhaye toh specific measure mein problem hai.*

**Q7
7****Manager chahta hai ki sirf Top 10 Products dikhein chart mein — kaise karoge?**

Teen approaches hain. Pehla: Visual filters use karo — chart select karo → Filters pane → ProductName → Filter type: Top N → 10 → By value: [Total Revenue] → Apply. Yeh sabse simple hai. Doosra: DAX mein TOPN() function use karo. Teesra: RANKX() se rank column banao phir <=10 filter karo. Visual-level Top N filter sabse easy aur recommended approach hai beginners ke liye.

■ *Always use Top N filter for dashboards — yeh dynamically update hota hai jab data refresh hota hai.*

**Q7
8****Month name sort order galat hai — January, February order mein nahi aa raha — kaise fix karo?**

Yeh common problem hai. Month name text hai — Power BI alphabetically sort karta hai — April, August, December... Correct sort: MonthName column ke sort column ko MonthNo (1-12) pe set karo. Data View mein MonthName column select karo → Column Tools tab → Sort by Column → MonthNo. Ab MonthName January, February, March order mein sort hoga. Date_Dim mein MonthNo (1-12 numeric) column hona chahiye isliye.

■ *Yahi issue DayName ke saath bhi hota hai — DayName ko DayOfWeek (1-7) se sort karo.*

**Q7
9****Ek column mein numbers aur text dono hain — kaise handle karo?**

Mixed data type column problematic hota hai. Power Query mein pehle column ka type 'Text' set karo. Phir numbers extract karne ke liye: (1) Replace non-numeric characters. (2) Custom Column banao: M code: = try Number.FromText([Column]) otherwise null. Numbers milenge text cells null honge. Ya dono alag columns mein extract karo — ek numeric ke liye ek text ke liye. Phir apne use case ke hisaab se jo chahiye woh use karo. Data cleaning is key.

■ *Best practice: Source data se pehle fix karwao agar possible ho — Power BI mein cleaning workaround hai, permanent solution nahi.*

**Q8
0****YTD correctly nahi calculate ho raha — December mein bhi reset ho raha hai mid-year — kya problem hai?**

Yeh fiscal year issue hai. TOTALYTD default January 1 se reset karta hai. Agar company ka fiscal year April 1 se start hota hai — formula mein year_end_date argument add karo: TOTALYTD([Revenue], 'Date'[Date], '3/31'). '3/31' matlab March 31 — fiscal year end. Ab April 1 pe reset hoga, January mein nahi. Date_Dim mein FiscalYear column bhi banana helpful hota hai filtering ke liye.

■ *Always clarify with client: 'Calendar year use karein ya Fiscal year?' Before writing Time Intelligence measures.*

**Q8
1****Multiple currency mein data hai — kaise handle karte hain Power BI mein?**

Approach 1: Exchange rate table banao — Currency, Date, Rate columns. Fact table se join karo. Measure: Revenue in INR = SUMX(Sales, Sales[Revenue] * RELATED(ExchangeRates[Rate])). Approach 2: Parameter ya slicer se currency select karne do — SWITCH(SELECTEDVALUE(Currency[CurrencyName]), 'USD', [Revenue]*83, 'EUR', [Revenue]*90, [Revenue]). Ek base currency choose karo (INR) aur sab convert karo.

■ *Dynamic currency conversion real-world projects mein common hai — yeh concept explain karna impressive hota hai interview mein.*

**Q8
2****Dashboard ko mobile ke liye kaise optimize karte hain?**

View tab → Mobile Layout mein jaao. Separate canvas dikhega — phone screen ke size ka. Yahan visuals drag-drop karo phone ke liye optimized layout mein. Desktop aur Mobile layouts alag save hote hain — desktop report affect nahi hoti. Tips: Mobile mein 2-3 most important visuals rakho. Touch-friendly slicers use karo. Font size larger rakho. Tile slicer se tap-friendly buttons. Power BI Mobile app pe preview karo publish karne se pehle.

■ *Best practice: KPI Cards sabse upar, phir ek main chart neeche — mobile mein scroll natural hota hai.*

**Q8
3****Ek table mein 10 million rows hain — kya Import Mode use karen?**

10 million rows Import Mode mein possible hai — Power BI efficiently compress karta hai (Vertipaq engine). Lekin refresh time zyada lagna chahiye aur file size badi hogi. Steps: Power Query mein unnecessary columns remove karo. Date ranges filter karo — purana irrelevant data exclude karo. Aggregation tables consider karo. 50 million+ rows pe DirectQuery ya Composite Model consider karo. Test karke dekho — performance acceptable hai ya nahi decide karo.

■ *Rule of thumb: 0-5M rows → Import confidently. 5-50M → Import with optimization. 50M+ → DirectQuery ya Composite.*

**Q8
4****Report mein alag-alag pages ke liye alag-alag RLS roles chahiye — kaise karen?**

RLS report level pe lagta hai — page level nahi. Ek hi role sab pages pe apply hota hai. Agar different users different pages dekhein → alag reports banao aur alag roles assign karo. Ya page visibility bookmarks se control karo — user role ke hisaab se relevant bookmarks dikhao. Enterprise mein alag workspaces ya apps use karte hain different user groups ke liye. Power BI Apps se different audiences ko curated content deliver kar sakte hain.

■ *Interview point: 'RLS data restrict karta hai — page visibility UI level pe hoti hai, separate concerns hain.'*

**Q8
5****Sales current month aur last month compare karna hai — kaise karoge?**

Teen measures banao: Current Month = CALCULATE([Total Revenue], DATESMTD('Date'[Date])). Last Month = CALCULATE([Total Revenue], PREVIOUSMONTH('Date'[Date])). MoM Change % = DIVIDE([Current Month] - [Last Month], [Last Month], BLANK()). Ek card mein Current Month dikhao, doosre mein Last Month, teesre mein change %. KPI visual mein Indicator = Current Month, Target = Last Month lagao — automatically red/green arrow dikhega.

■ *Pro tip: Conditional formatting add karo change % card pe — negative values red, positive green. Very professional look!*

**Q8
6****Aapke paas Excel file aur SQL database dono se data lena hai — kaise combine karoge?**

Power BI dono sources ek saath handle kar sakta hai. Get Data → Excel (pehli file load karo). Get Data → SQL Server (database connect karo). Dono queries Power Query mein separately aayenge. Agar combine karna hai: Append (same structure) ya Merge (join ke liye). Ya Model View mein relationships banao dono tables ke beech agar common key hai. Har source ke credentials alag hote hain — Service mein refresh ke liye dono ka gateway configure karna padta hai.

■ *Real-world common scenario: HR data Excel mein, Sales data SQL mein — Power BI dono ko ek report mein easily combine karta hai.*

**Q8
7****Report mein conditional title banana hai jo filters ke saath change ho — kaise?**

Measure banao: Report Title = 'Sales by ' & SELECTEDVALUE(Products[Category], 'All Categories') & ' | ' & SELECTEDVALUE('Date'[Year], 'All Years'). Phir visual select karo → Format pane → Title → Title text ka 'fx' button → Field value → us measure select karo. Ab jab slicer change karo — title automatically update hoga. Yeh very professional touch hai dashboards mein — users clearly samajh te hain kya dekh rahe hain.

■ *Extend karo: Page title bhi dynamic banao using text box with measure — entire experience contextual hoti hai.*

**Q8
8****Client chahta hai ki previous 12 months ka data hamesha dikhe regardless of date slicer — kaise?**

DATEADD use karo: Last 12 Months = CALCULATE([Total Revenue], DATEADD('Date'[Date], -365, DAY)). Ya better: CALCULATE([Total Revenue], FILTER(ALL('Date'), 'Date'[Date] >= TODAY()-365 && 'Date'[Date] <= TODAY())). Yeh slicer se independent hoga. Ya visual level filter use karo — Date column pe 'is in the last 12 months filter lagao — yeh slicer override karta hai. Har approach ke pros/cons explain karo.

■ *Client requirement: 'Rolling 12 months' — yeh common business requirement hai. TODAY() function dynamically current date use karta hai.*

**Q8
9****Power BI report ko PowerPoint mein export karna chahte hain — kaise karte hain?**

Power BI Service mein: File → Export → PowerPoint. Yeh report ke snapshots ko PowerPoint slides mein convert karta hai — har page ek slide banta hai. Alternatively: Export to PDF — har page PDF page banta hai. Live interactive export nahi hota — sirf static images. Agar live interactive content chahiye → Embed report in PowerPoint plugin available hai — slides mein interactive Power BI embed kar sakte hain. Desktop se direct export nahi hota — pehle Service pe publish karna zaroori hai.

■ *Client presentations ke liye Export to PowerPoint most used feature hai — quick professional presentation ready ho jaati hai.*

**Q9
0****Aapko ek project mein galti ho gayi — wrong data model banaya. Kaise discover kiya aur fix kiya?**

Honestly batao — 'Maine relationships galat direction mein banaye the — Fact table se Dimension table ki taraf. Issse filter context ulta flow kar raha tha — Region filter lagane pe sirf woh region ka data nahi aa raha tha saari regions ka aa raha tha.' Discovery: Results unexpected the — cross-check kiya Excel pivot table se. Fix: Model View mein relationships dobara create kiye correct direction mein. Learning: Ab hamesha pehle test data se verify karta hoon relationships banane ke baad.

■ *Interview tip: Galti acknowledge karo honestly — recovery aur learning zyada important hai. 'Maine seekha ki relationships validate karna zaroori hai.'*

Section 6 — HR, Tips & Best Practices

Q91–Q100 | Final round aur behavioral questions

**Q9
1****Power BI aur Tableau mein kya difference hai?**

Power BI Microsoft ecosystem mein best fit karta hai — Excel, Azure, Teams ke saath seamless integration. Cost-effective — Pro license mein reasonable pricing. DAX powerful language hai complex calculations ke liye. Tableau visually more polished hai aur non-technical users ke liye easier drag-drop experience. Tableau SQL server ke saath better tha historically lekin Power BI rapidly improving hai. Power BI market mein Gartner Magic Quadrant mein leader hai — India mein bahut popular hai especially Microsoft shops mein.

■ *Smart answer: 'Dono strong tools hain — Power BI Microsoft stack ke liye best, Tableau cross-platform ke liye. Job requirement dekho — Power BI India mein zyada demand hai.'*

**Q9
2****Apna sabse challenging Power BI project batao.**

Structure: Situation → Task → Action → Result. 'Ek sales dashboard project mein client ke paas 5 alag Excel files thi alag formats mein. Challenge: Sab combine karna aur consistent data model banana. Maine Power Query mein custom functions banaye jo automatically saari files load karti thi. Data model mein Star Schema apply kiya. DAX mein Time Intelligence measures banaye. Result: Client ko ek click mein updated dashboard milti thi jo pehle 4 ghante ka manual work tha.' STAR method use karo.

■ *Agar real project nahi hai: Capstone project ya course project describe karo with same STAR structure — bilkul honest keep karo.*

**Q9
3****Power BI mein aap apne aap ko kahan rate karte hain (1-10)?**

Honest self-assessment do. Beginner freshers ke liye: '6 — Power Query, Star Schema, basic DAX, aur 3-4 visual types mein comfortable hoon. Time Intelligence aur advanced DAX pe aur practice chahiye. Har din seekh raha hoon.' Overconfident mat lagao — HR notice karta hai. Undersell bhi mat karo. Specific strengths aur improvement areas dono mention karo. Yeh question growth mindset test karta hai.

■ *Best response format: 'Rating X/10. Strong in A, B, C. Improving in D, E. Currently working on F.'*

Q9
4**Power BI mein latest features kya aaye hain jo aapko pata hain?**

Power BI monthly updates aate hain — Microsoft blog follow karna zaroori hai. Recent common updates: (1) Copilot integration — AI se reports generate aur insights. (2) Paginated Reports improvements. (3) New visual types. (4) Enhanced Q&A.; (5) Power BI Goals/Scorecards. Batao ki 'Main Microsoft Power BI Blog regularly follow karta hoon aur monthly release notes padhta hoon.' Specific feature example do latest months ka.

■ *Interview tip: powerbi.microsoft.com/en-us/blog pe monthly updates aate hain. Ek-do latest features note karke interview mein mention karo — initiative impressive hoti hai.*

Q9
5**Measure vs Calculated Column — interview mein demonstrate karke dikhao.**

Concrete example do: 'Orders table mein Profit column banana tha — har order ke liye Revenue minus Cost. Yeh row-level value hai toh Calculated Column banaya: Profit = Orders[Revenue] - Orders[Cost]. Ab Total Profit report mein chahiye — yeh aggregate hai toh Measure banaya: Total Profit = SUM(Orders[Profit]). Calculated Column file mein permanently stored hai — 300 rows ke liye 300 values. Measure dynamically calculate hota hai — visual filter ke hisaab se.' Practical example clearly demonstrate karta hai understanding.

■ *Strong answer structure: Definition → When to use → Concrete example → Code — yahi sequence impress karta hai.*

Q9
6**Aapne galat measure banaya jiska result expected se alag tha — kaise debug kiya?**

Debug process batao: 'DAX measure tha jo product-wise margin dikhana tha. Matrix visual mein galat values aa rahi thi. Pehle simple Card visual pe same measure lagaya — correct total tha. Phir Table visual pe ProductName ke saath lagaya — kuch products mein value galat thi. Discover kiya ki RELATED() wrong table reference kar raha tha. Products_Dim mein do similar columns the — wrong ek select ho gaya tha. Fix kiya — results correct aa gaye. Learning: Measure hamesha isolated card pe pehle test karo, phir complex visual pe.'

■ *Debugging steps share karna technical maturity dikhata hai — interviewers appreciate karte hain structured problem solving.*

Q9
7**Dashboard design ke 5 best practices kya hain?**

Pehla: 5-second rule — koi bhi user 5 seconds mein samajh jaaye ki dashboard kya bol raha hai. Doosra: KPI cards top-left se top-right arrange karo — most important pehle. Teesra: Consistent color theme — max 3-4 colors, random colors mat use karo. Chautha: Har visual ka clear meaningful title. Paanchwa: White space use karo — crowded dashboard overwhelming lagta hai. Bonus: Mobile layout bhi banao — aaj kal mobile access important hai.

■ *One line summary: 'Good dashboard = right data + right visual + right layout + right user.' Yeh ek sentence interview mein great impression banata hai.*

Q9
8**Power BI mein data refresh kaise schedule karte hain?**

Service pe publish karne ke baad: Workspace mein Datasets section jao → us dataset ke '...' menu → Settings → Scheduled Refresh. Gateway configure karo agar on-premise data source hai — Power BI Gateway install karna padta hai local machine ya server pe. Refresh frequency set karo — Pro license mein max 8 per day, Premium mein 48 per day. Refresh failure pe email notification ON karo. Cloud sources jaise SharePoint Online, Azure ke liye gateway ki zaroorat nahi.

■ *Common interview question: 'Gateway kab chahiye?' Answer: On-premise sources (SQL Server, local Excel files) ke liye gateway zaroori hai. Cloud sources ke liye nahi.*

Q9
9**Aap Power BI mein kaise seekhte rehte ho — resources kya use karte ho?**

Honestly batao: Microsoft Documentation (docs.microsoft.com/power-bi) — official aur updated. SQLBI.com — DAX ke liye best resource, Marco Russo aur Alberto Ferrari ke articles. Guy in a Cube YouTube channel — weekly Power BI tips. Microsoft Learn — free structured learning. Power BI Community forums — problems solve karne ke liye. Kaggle aur real datasets pe practice karna. Monthly release notes padhna habit banao. Courses complete kiye — certificates relevant hain.

■ *Show curiosity: 'Main har week 30-60 minutes Power BI practice karta hoon nayi features try karne ke liye.' Consistent learning discipline impressive hoti hai.*

**Q1
00****Ek interviewer aapko bolta hai 'Hume ek Sales Dashboard banana hai — pehla step kya hoga aapka?'**

Structured answer do: 'Pehle requirements gather karunga — audience kaun hai (CEO, Sales Manager, Operations)? Kaunse KPIs important hain unhe? Data sources kaun se hain? Report ka frequency kya hoga — daily, weekly, monthly? Phir data explore karunga — quality check, missing values, relationships. Power Query mein clean karunga. Star Schema banaunga. Basic measures likhke validate karunga. Phir ek wireframe sketch karunga pages ka. Client ke saath approve karake phir visuals build karunga. Finally user testing aur feedback loop.' Poora process batana maturity dikhata hai.

■ *Best interview answer: Jo sirf 'Maine Power BI khola aur banana shuru kiya' nahi bolta — systematic approach dikhata hai. Requirements → Data → Model → DAX → Visuals → Test — yahi professional approach hai.*

All The Best! ■

Interview mein confidence ke saath jao — aap prepared ho!

- ✓ Interview se pehle raat ko yeh 100 questions ek baar scan karo — sirf headlines padho.
- ✓ Sabse important questions: Q1, Q6, Q7, Q9, Q10, Q20, Q21, Q22 — yeh hamesha puche jaate hain.
- ✓ Har answer mein ek real-life example do — 'Maine ek project mein yeh kiya tha' wala angle.
- ✓ Agar answer nahi pata toh honestly bolo: 'Mujhe exactly recall nahi aata, but approach yeh hogi...'
- ✓ STAR Method use karo behavioral questions ke liye: Situation, Task, Action, Result.
- ✓ Power BI ke baare mein passionate lago — enthusiasm interviewers notice karte hain.
- ✓ Salary negotiation: Market research karo pehle — Glassdoor, Naukri pe check karo.
- ✓ Final question hamesha poocho: 'Is role mein typical day kaisa hota hai?' — interest dikhata hai.

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